Statistics - 30001

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Dept. Decision Sciences

Course Introduction: Objectives

PART I
DESCRIPTIVE

PART II
INFERENCE

The course introduces:

- Techniques to properly **describe** and **summarize** different types of data, and to study their **relationships**
- **Sampling** and **statistical inference**, with a focus on the potential **risks** associated to inference. The course assumes that students have a basic understanding of **probability** theory and **random variables**, which are covered in the course Mathematics Module 2 (Applied) - code 30063.

The use of these tools in practice is illustrated through applications based both on **aggregated data** and on **datasets**. In the latter case, we will use the R statistical software and in particular the integrated development environment **RStudio**. Besides some basic functions, we will use the package **UBStats** developed exclusively for the course. The course heavily relies on the software, that will be also used during the exam. It is therefore essential to have both R and RStudio installed on your laptops both to attend the lessons and to take the exam.

Course Introduction: Lectures

The course is structured as follows:

- 28 lectures: introduction of the fundamental concepts of statistical data analysis and presentation of the main functions available in RStudio.
- 10 practical sessions (scheduled once per week): application of the techniques taught in the lectures by presenting problems to be solved either manually and/or using R tool.
- Online meetings supporting students' self-study; Q&A on exercises assigned on a weekly basis and on the application of the topics introduced during lectures and practical sessions. The schedule of the meetings will be posted on Bboard page common to all the classes **(30001 statistics - material exams)**

Teaching materials

- Materials common to all the classes, reference slides, specific manual on the R software, summary exercises, mock exams, past exams, Q&A, available on the Bboard page common to all the classes (**30001 statistics - material exams)**
- WE HAVE A BOOK TAILORED TO THIS COURSE!

Course Introduction: Exams

All the students can take the exam in two ways:

- 2 midterms (1 hour; graded with 31/30 points maximum). Each midterm is passed with a grade of 15 or higher. If both midterms are passed, the final grade is the average of the two achieved grades. To pass the overall exam, the final grade must be at least 18.
- 1 general exam (2 hours; graded with 31/30 points maximum). The exam is passed with a grade of 18 or higher

<u>All</u> exams are closed book (you can use the R manual only) and involve

- Questions on theoretical aspects (definitions, proofs, properties of statistical tools)
- Traditional exercises based on summaries of data (to solve manually - calculator or R)
- Questions related to the analysis of data to be answered using Rstudio, which should be installed **on your laptop** before the exam (Please be sure that the software is properly set up and functioning ahead of time)

Students attending at least the 75% of the lessons both in the first and in the second part of the course and who take the exam in **January or February** will get 1 point added to their final grade (Please note that such additional point does not contribute to the laude)

You need to register within 30 mins from start of class

Workflow suggested by Pier

- Just **before class read the** (non annotated) **slides** to anticipate topics
- Attend class! And during class try and be focused, engage and ask questions, make the most of your time – which is super precious!
- After class revise the annotated slides, write down doubts. Try and clarify them with your peers or ask me in the next lecture or via email.
- Try and replicate the examples, exercises and scripts.
- Attempt – without looking at the solution first! – the available exercises from our Bboard and from the common Bboard.

Main Resources for our specific adventure (Pier)

- Annotated slides (posted after each class)
- Exercises sheets with solutions
- Scripts from class (posted after each class)
- R Videos
- Past papers in the common BBoard
- Solved exercises in the common BBoard

Slides color coding – specific to our class (Pier)

- **Blue slides** are fundamental concepts or information
- **Red slides** are related to applications and R Studio
- **Green slides** are **not** optional, but imo it's enough to read through them. They might help your intuition.
- **Gray slides** are **optional.** They might be interesting in depth analysis, but you'll survive without them.

EXTRA ADVICE

- Part II is more difficult than Part I

→ during part I do a lot of practice with R Studio!

MY GOALS

1. EXAM
2. PREPARE YOU FOR LIFE AFTER 30001
3. HAVE FUN

STATISTICS: BASIC CONCEPTS

Decision making under uncertainty

In many cases it is necessary to make decisions, evaluate scenarios, define business strategies or political actions under uncertainty.

- Identify the buying habits of a company's customers
- Determine the most effective marketing strategy for a certain target audience
- Identify the most profitable customers
- Select the candidates to admit to a master's program
- Forecast the amount of orders for the coming year
- Assessing the efficacy of a vaccine
- Determine whether the average wages of men and women are different (gender pay gap)
- stimating the rate of tax evasion

In all these cases, proper statistical analysis of the data can support decision making.

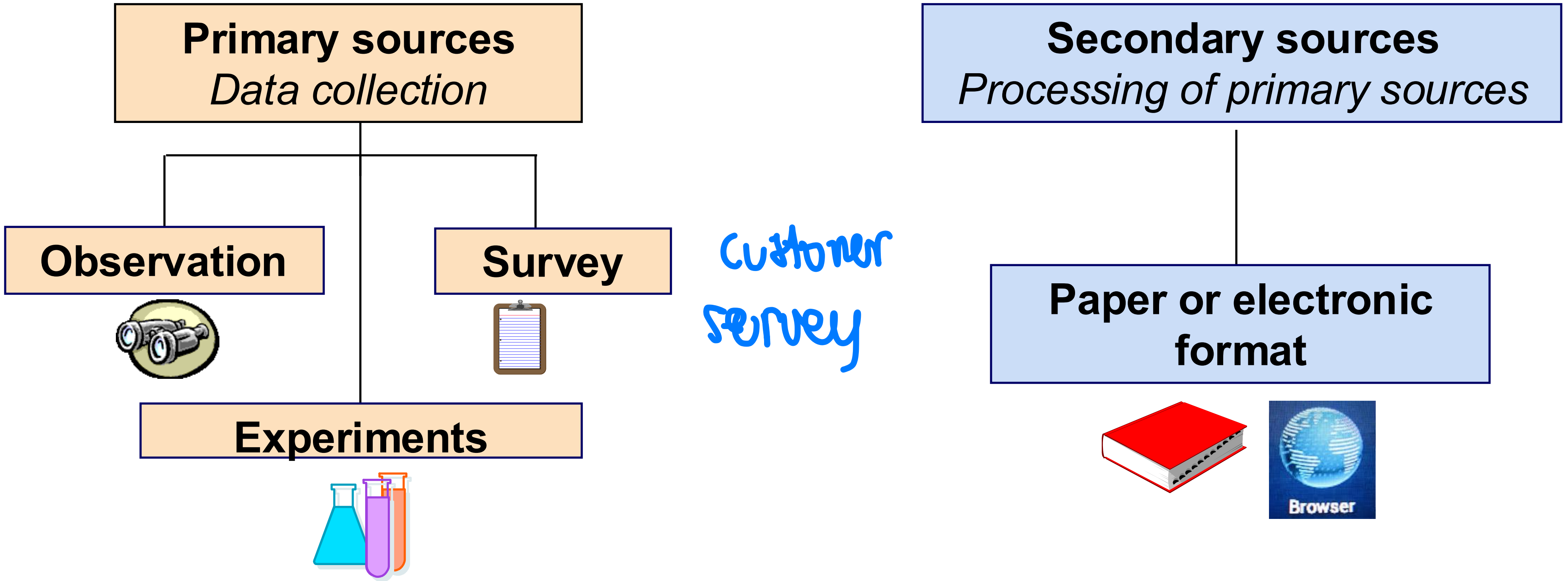
Decision making under uncertainty

The statistical method encompasses all the necessary steps and procedures for processing and analyzing data to extract valuable and immediately applicable information, potentially aiding decision-making processes.

1 - **Data collection and organization**
2 - **Processing and analyzing data to describe and summarize their characteristics**
- **Accurate and effective result description, communication and interpretation**
3 - **Possible extrapolation from the obtained information, with a proper assessment of the associated risks**

Data collection and data preparation [not done in 30001]

A first, fundamental step is to define the required data concerning a specific problem, either by collecting new data or identifying existing information sources.



Company accounts

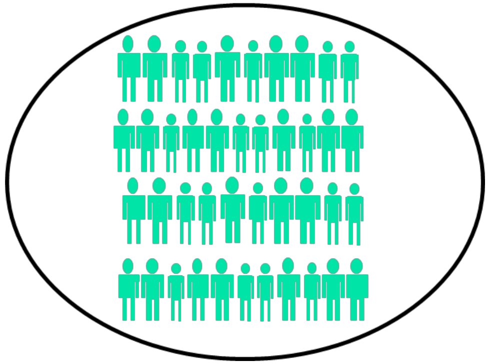
customer survey

medical statistics

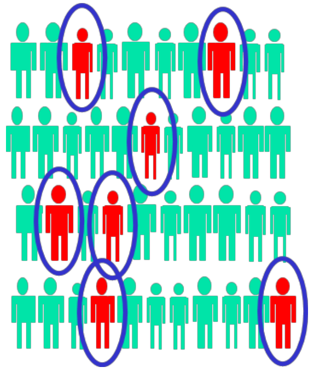
Population and sample

Data may refer to a **population** or a **sample**

- **Population** (universe): the complete set of **all** units (***N***) of interest in a study (all EU countries, all the citizens in a country, all the customers of a company,…).



- **Sample**: ***subset of*** the **population**. In some cases, collecting data on the entire population may be prohibitively expensive (high cost) or difficult or even impossible. For example, when the population is very large (set of all companies operating in Europe) or not fully accessible (difficulty in identifying all individuals in the population, e.g., all customers who purchased products of a certain brand); gathering data on the entire population becomes unfeasible.
  In such cases, it becomes necessary or convenient to collect data only on a portion of the population, which is represented by a sample (***n***)



A/B testing

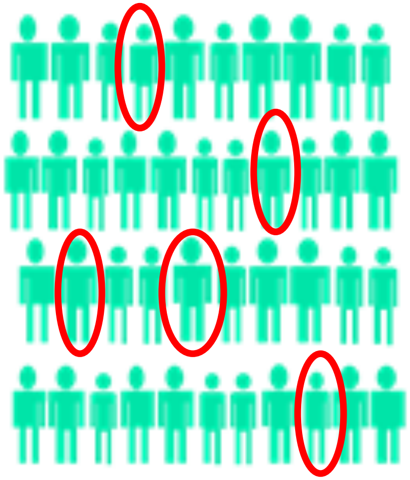
Random samples ↳ BIASED SAMPLE wrt age, education, region...

e.g. a sample of friends at Bocconi is NOT REPRESENTATIVE OF THE POPULATION

To ensure the representativeness of the samples, it is essential to draw them **randomly.** This way, the samples accurately reflect all units in the population and avoid biases towards specific groups of subjects

Simple random sampling. The sample is selected in such a way that

- Each unit is randomly and independently selected as a sample unit
- Each unit of the population has an equal chance of being chosen
- All the possible samples of the same size n have the same chance of being chosen



$N = 42$ $n = 5$

There are more sophisticated sampling schemes and methods, that are not covered in the course. The focus will be on understanding and using simple random samples.

in 30001 we'll assume DATASETS are RANDOM SAMPLE from POPULATION

Key definitions: observations, variables and data

Observation, case, or statistical unit: entity (population unit) on which information is collected

Variable: a characteristic of interest related to the cases under study

Data (or measurements): observations made about the variable of interest measured on the cases under consideration

Values (or levels): distinct values taken by the variable

Data are commonly arranged in datasets, with **cases** listed in rows and **variables** represented in columns.

Datasets: examples

Ease of doing business:

survey of the ease of doing business in different countries worldwide

Information on **all countries globally**

- **Cases:** the countries (observations, statistical units)
- **Variables**: indicators related to the ease of doing business in each country:
  - Starting or closing a business, paperwork requirements for building permits and property registration, labor regulation, access to credit, investor protection, taxation, trade, contract enforcement procedures...
- **Data**: measurements on these indicators observed for each individual country

Datasets: examples

Collected data in the Ease of doing business survey (first 25 rows)

variables

statistical units / observations

1	CODE	Starting	Constructions	Employment	Property	Credit	Investors	Taxes	Trade	Contracts	Closing
2	AFG	7.65	20.34	24.84	45.02	2.5	5	6.84	63.69	58.91	100
3	AGO	23.15	11.66	61.71	50.85	27.55	58.06	14.99	57.72	49.25	59.2
4	ALB	8.5	19.85	34.17	19.48	41.88	75.83	20.44	22.84	29.4	100
5	ARE	12.35	12.6	17.06	7.77	34.89	44.44	6.37	12.78	40.88	58.87
6	ARG	20.08	22.66	36.92	22.38	70.21	46.67	23.52	27.25	26.28	35.82
7	ARM	11.36	11.01	29.03	5.62	45.28	50	22.92	31.63	32.03	24.81
8	ATG	10.6	7.8	11.06	27.36	17.5	65.28	17.8	16.8	30.61	32.23
9	AUS	1.4	11.43	2.94	16.29	68.33	56.94	13.59	17.33	16.18	10.13
10	AUT	10.78	8.73	31.44	12.44	53.15	40.56	18.43	11.66	13.03	17.55
11	AZE	7.3	21.79	3.75	8.64	41.85	68.06	16.09	57.04	22.18	33.58
12	BDI	27.02	22.76	29.76	29	9.26	32.5	41.92	52.88	42.7	100
13	BEL	3.15	8.82	19.98	38.98	53.05	71.67	20.34	15.09	14.93	5.3
14	BEN	22.22	15.42	37.54	29.56	15.1	32.5	25.97	29.05	46.14	49.15
15	BFA	12.02	10.88	20.44	33.7	12.29	36.11	18.68	53.53	42.82	41.57
16	BGD	11.97	10.68	35.32	46.18	28.63	68.61	10.11	25.61	57.44	40.58
17	BGR	6.07	15.28	28.15	21.8	56.3	60	16.12	23.8	29.87	35.39
18	BHR	9.42	5.42	21	5.53	35.62	57.5	8.38	14.65	37.34	21.85
19	BHS	9.54	12.37	15.67	33.27	22.5	47.5	9.44	18.2	36.92	30.93
20	BIH	18.58	13.58	42.44	29.79	50.63	51.11	19.53	19.19	32.61	34.02
21	BLR	10.81	11.87	25.82	8.93	26.62	46.39	47.49	30.02	13.29	49.38
22	BLZ	15.01	4.25	14.02	27.33	20	43.61	11.85	27.17	47.78	22.35
23	BOL	26.66	12.85	88.25	27.09	38.82	40.56	25.47	27.22	33.22	30.98
24	BRA	28.29	17.39	43.9	39.16	50.49	54.72	35.72	23.66	34.7	44.56
25	BRN	27.05	21.24	6.95	.	17.5	43.06	9.29	18.23	48.67	24.95

measurements

Datasets: examples

Loyalty cards and interviews

Loyalty cards offer companies a wealth of information about customer characteristics and behaviors:

- Upon subscription, data is collected, including gender, age, and area of residence. The loyalty card continues to provide valuable information, such as the number of visits, amount spent during a specific period, customer profitability (calculated based on spending, discounts, and marketing costs), and details of their most recent purchase.
- Additionally, a specific targeted analysis can be carried out on a **random sample of** customers to gather information about their satisfaction levels regarding store features, value for money, and their propensity to recommend the service.

Cases: the selected clients (the sample available for the analysis)

Variables: information gathered from the loyalty card records and from interviews to clients

Data: actual values on the variables measured on clients within the selected sample

Datasets: examples

QUANT. CONTINUOUS DATA COLLECTED IN CLASSES

Dataset (first 15 cases) created by merging information obtained directly from customers, data collected through loyalty card records, and data gathered through interviews conducted with a selected sample of customers

Customer	Age	Income	Payment	Dept	Nr_visits	Amount	Profitability	Recommend
1	35	[5-15)	Credit Card	Clothing	4	263.2	81.8	Neutral
2	25	[15,20)	Cash	Clothing	4	243.2	82.6	Neutral
3	54	[20,25)	ATM	Clothing	3	273.6	85.7	Neutral
4	60	[25,35)	Cash	Clothing	7	585.2	86.8	Neutral
5	44	[35,60)	ATM	Clothing	2	151.6	88.9	Very Unlikely
6	62	[15,20)	Credit Card	Clothing	2	102	91.5	Neutral
7		[35,60)	Cash	Accessories	1	84.6	91.7	Unlikely
8	55	[20,25)	Cash	Accessories	3	276.75	92.8	Very Unlikely
9	62	[20,25)	Credit Card	Accessories	2	185.4	94.3	Very Unlikely
10	64	[20,25)	Cash	Accessories	2	207	96.2	Likely
11	55	[25,35)	Cash	Perfumery	3	191.7	103.8	Likely
12	58	[25,35)	Cash	Kids	4	195.6	104.2	Very Likely
13	49	[15,20)	Cash	Home/Textiles	3	160.65	105	Unlikely
14	59	[15,20)	Cash	Home/Textiles	2	104.4	108.6	Unlikely
15	70	[5-15)	ATM	Home/Textiles	2	100.2	110	Likely

The **values** of the surveyed variables have different **types** and characteristics!

QUAL. NOM. QUANT. DISC QUANT. DISCRETE QUANT. CONT.

Classification of variables

An initial classification of variables into **qualitative (categorical)** and **quantitative (numerical)** is based on the **type or nature** of the distinct values they take:

- The values (or levels) of **qualitative** variables are **labels** indicating attributes, **group** membership, or identifying specific **categories** with distinct characteristics (gender, region of residence, level of satisfaction).
- The values of **quantitative** variables are **numerical**

Classification of variables. Why do I care, Pier?

→ for each type of var we use different tools!

Qualitative and quantitative variables can be further classified into subcategories based on their characteristics:

Qualitative variables:

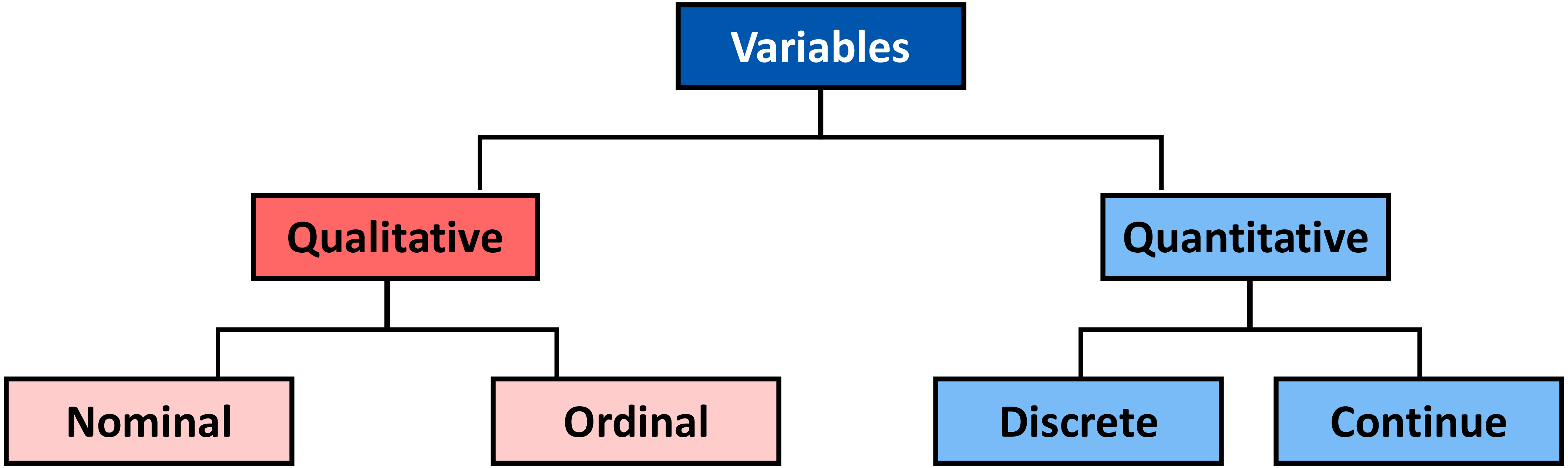
- **Nominal** (gender, business name of firms, region of residence): the variable's values **cannot be ordered**.
- **Ordinal** (satisfaction, level of education, positions in an organization chart, final ranking): the variables' values can **be ordered**, but their differences cannot be quantified

Quantitative variables:

- **Discrete** data are generated by a **counting** process (number of children, number of visits to a store): the variable's values are integers.
- **Continue** data are generated by a **measurement** process (amount spent, height of an individual, income): the values can take any real numerical value

Note: In some cases, numerical variables may be **collected in classes**. For example, when surveying respondents' personal income, it might be difficult for them to accurately state the precise amount, so the data might be collected in income intervals or ranges.

Data Types/Variables



Nominal

Examples:
- Gender
- Payment methods
- Region of residence

Categories
Cannot be ordered

Ordinal

Examples:
- Satisfaction levels
- Propensity to recommend
- Ranking

Categories
Can be ordered

Discrete

Examples:
- No. of purchased products
- No. of children

Numbers
Count

Continue

Examples:
- Amount spent
- Income
- Revenues

Numbers
Measurement

Parameters and statistics

In general, we are interested to measure specific **characteristics of a set of data**.

It is important to distinguish between:

population mean, population proportion

- **Parameter**: a measure that describes (or summarizes) a characteristic of the entire **population**. For example, we may be interested in assessing the proportion of Italian women who exit the labour market after the birth of their first child, or the proportion of European companies that registered a patent, or the average turnover of Italian companies operating in a certain industry.
- **Statistic**: a measure that describes (or summarizes) a characteristic of a **sample**. For example, the proportion of respondents satisfied with a certain service, or the average number of purchase transactions within a specific period, or the average amount spent by the individuals in a sample. While our primary interest is the value of the **parameter for the entire population**, we must rely on the value calculated from the available sampled data.

sample mean, sample proportion

POPULATION N $\xrightarrow{\text{RANDOM SAMPLING}}$ SAMPLE n

POPULATION $\downarrow$ DESCRIPTIVE STATISTICS

SAMPLE $\downarrow$ DESCRIPTIVE STATISTICS

POPULATION PARAMETERS $\xleftarrow{\text{INFERENTIAL STATISTICS}}$ SAMPLE STATISTICS

examples:

population proportion p

pop. mean μ

examples:

$\hat{p}$: sample prop.

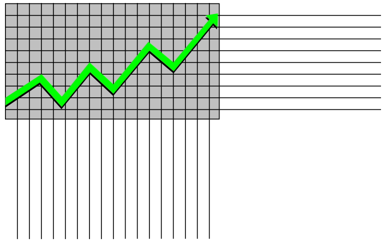
$\bar{x}$: sample mean

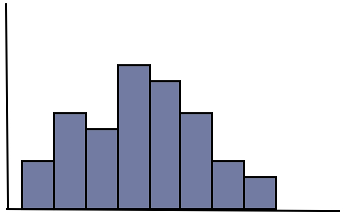
the GOAL of INFERENTIAL STATISTICS is to use SAMPLE STATISTICS to INFER on POPULATION PARAMETERS

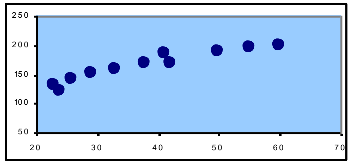
Descriptive statistics and inferential statistics

Descriptive statistics involves both graphical and numerical methods to summarize and analyze data. It can be applied to data from either the entire **population** or a **sample**. It includes a number of techniques for preparing, summarizing and effectively presenting data.

Data presentation: Tables and graphs







Data analysis and synthesis: Summary measures, such as mean, variance, or correlation

Descriptive statistics and inferential statistics

Inferential statistics involves methods used to make inferences and predictions about the population characteristics (**parameters**) based on information derived from sample data (**statistics**).

A crucial issue of this process is to evaluate the **reliability** of such extension and the **risks** associated with using **sample** information to draw conclusions about the **entire population**.

To accomplish this assessment, it is essential to take into account the **random mechanism** (**probability**) involved in the sample selection process, which leads to the inherent randomness in the fact that the available sample is just one of the many possible samples that could have been drawn from the population.

Bocconi

INTRODUCTION TO R & RSTUDIO

(Handbook on the use of R)

R tool: structure

R is a programming language designed specifically for data manipulation, statistical analysis and graphical representation.
It comes with an extensive collection of ***packages*** (sets of functions) that come pre-installed with the basic version of R or can be accessed after installation.

Data can be created from scratch in R, as well as imported from various sources (from hard disk or the web).

The results of data processing (data, graphs, functions) can be printed, or saved as objects in R. These objects can then be further analyzed, manipulated, or saved on hard disks for future use.

RStudio is an *integrated* development environment (IDE) for R. RStudio significantly simplifies the way users can interact with R and enhances the overall R programming experience.

Installation

To install R on your laptop, follow these instructions:

- Download from the R website (https://cran.r-project.org/) the executable file (.exe or .pkg) for the latest version of the R software for your operating system. For Windows, download from https://cran.r-project.org/bin/windows/base/ or for macOS download from https://cran.r-project.org/bin/macosx/base/) and run it.
- Follow the installation instructions, accepting all default options.

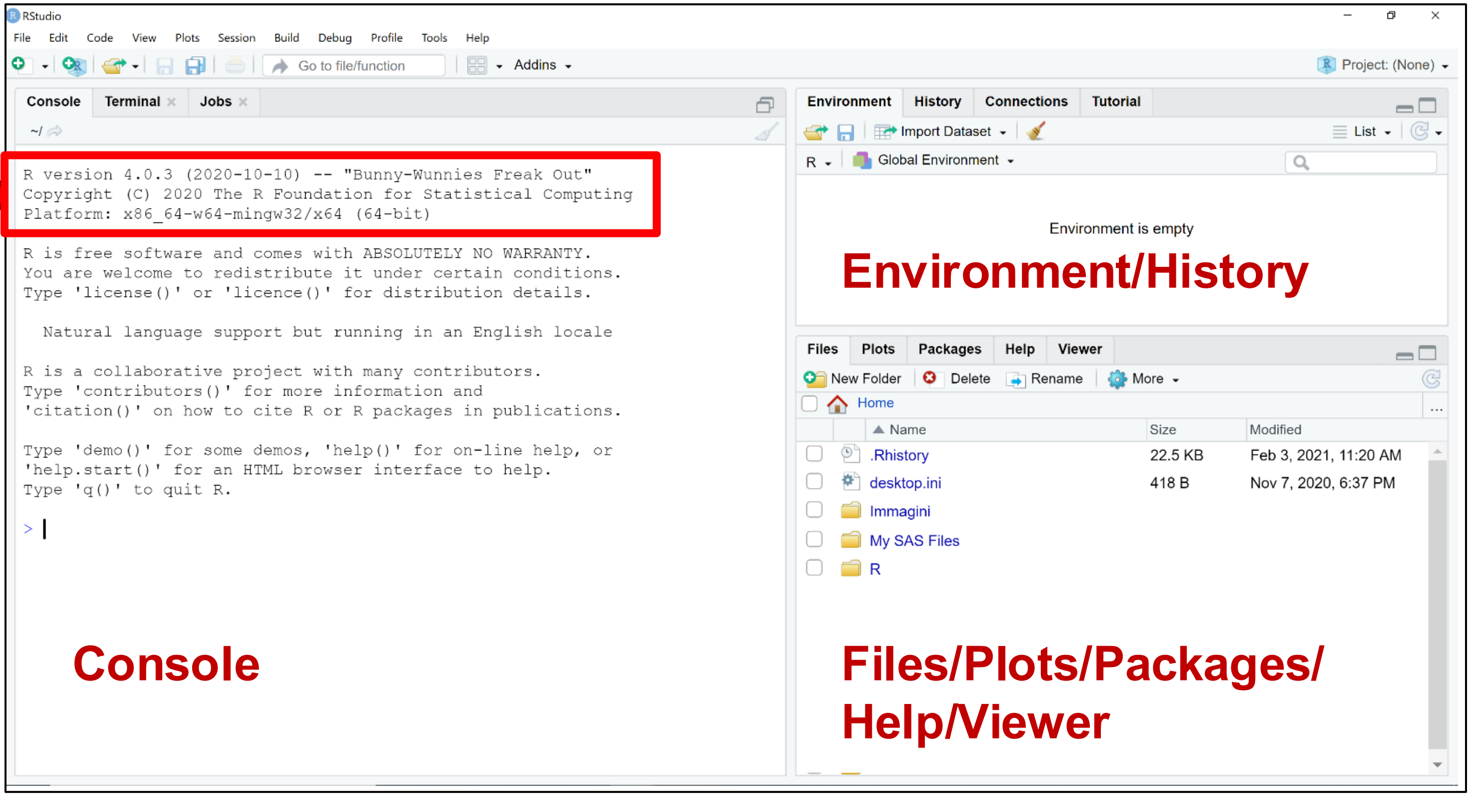
To install Rstudio, follow these instructions (also available on Bboard).

- Download from the RStudio website (https://rstudio.com/products/rstudio/download/) the executable file (.exe or .pkg) for the latest version of **Rstudio Desktop** and run it.
- Follow the installation instructions, accepting all default options.

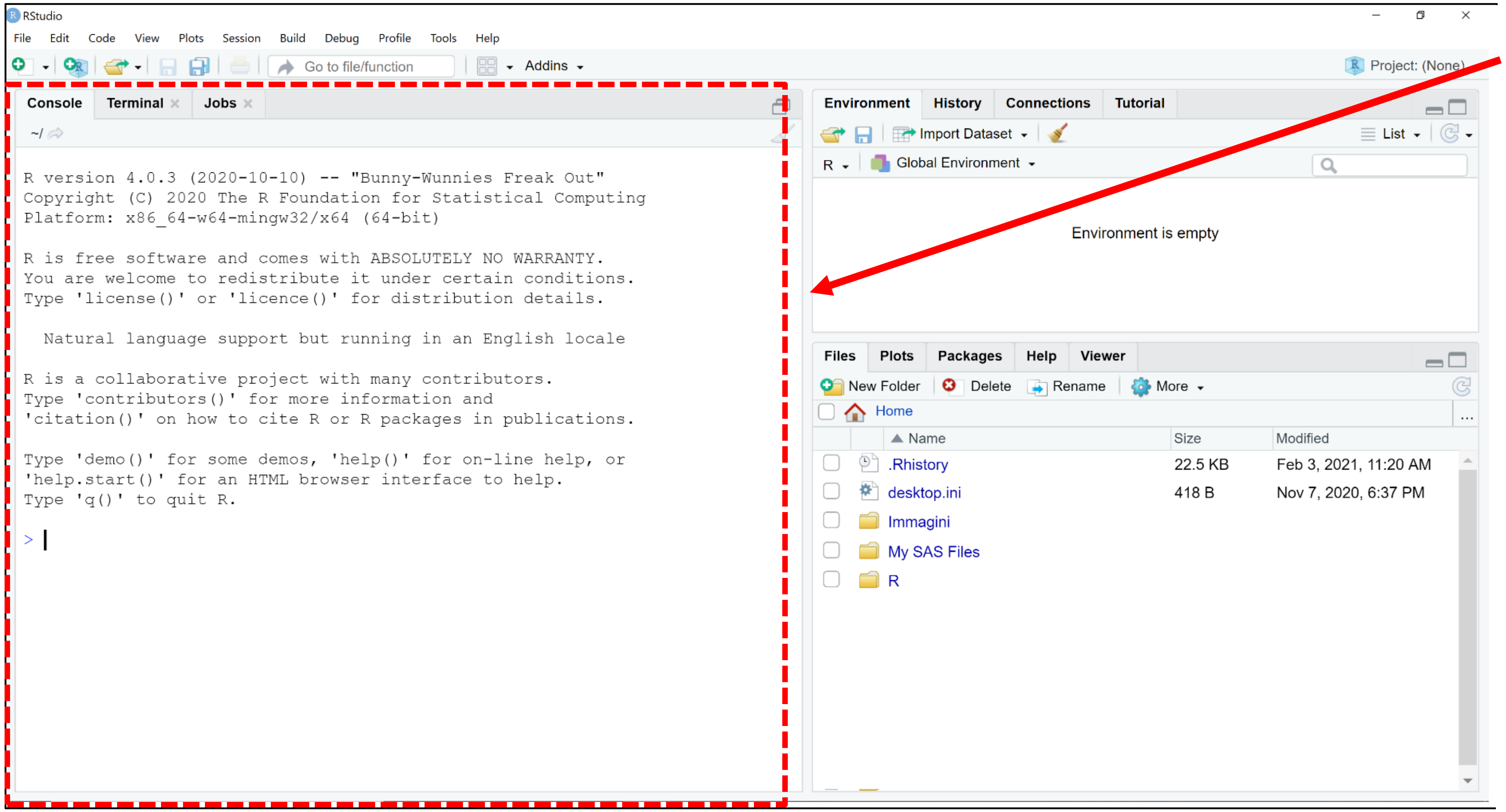
Panes in RStudio

When you start RStudio, three panes typically open:

Information about the R version being used (this is the previous version)

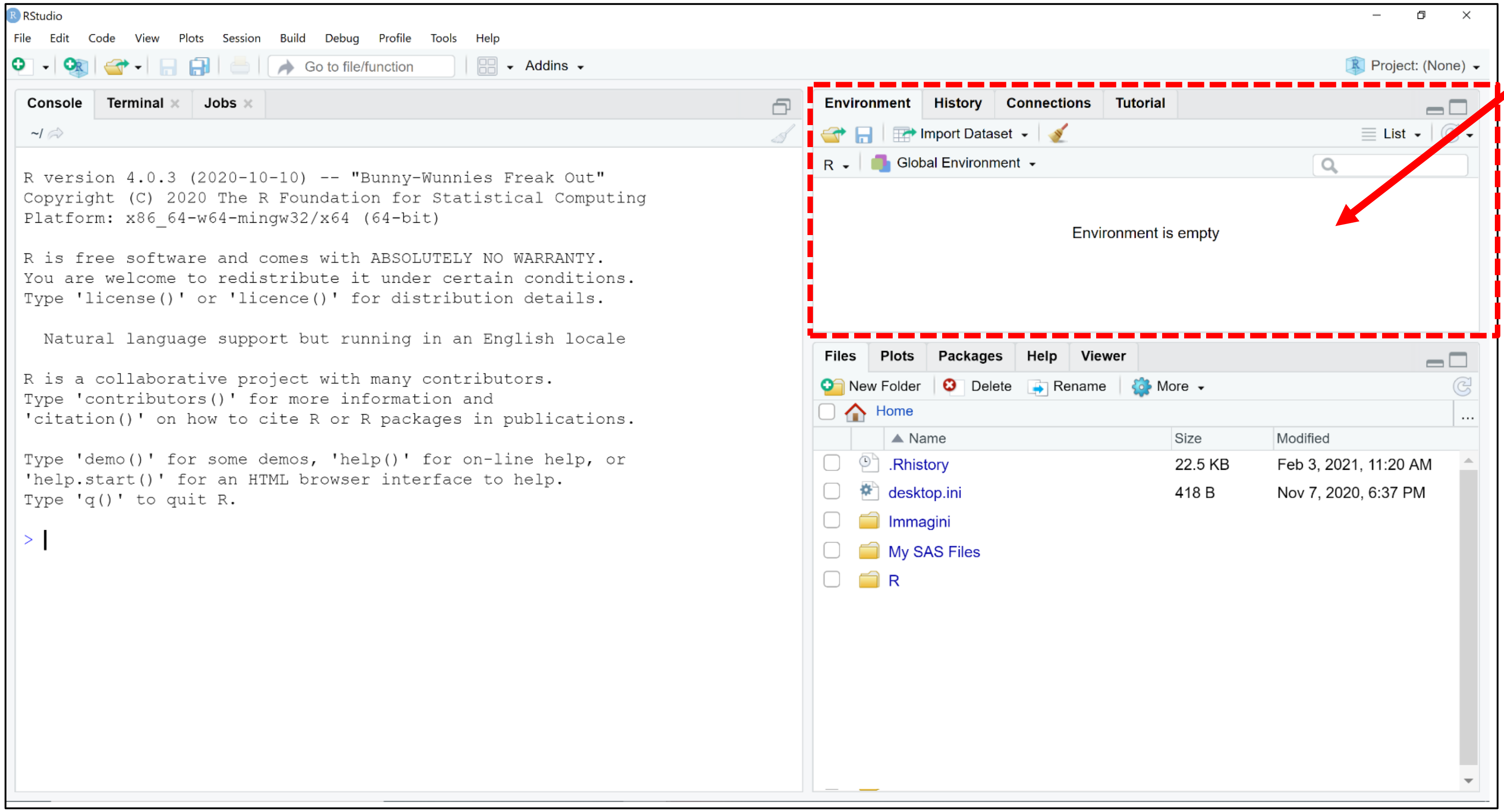


Panes in RStudio



Console: execution pane. It is the interactive part of RStudio, where you can directly type and execute R commands. It allows you to interact with R in real-time and to see the results of your commands immediately.
It reports submitted commands, provides information on the outcome of processing (reporting any errors), and, if required, displays the output. It is possible to work interactively and type commands directly into the console, but it is definitely better to collect them in a **script**

Panes in RStudio

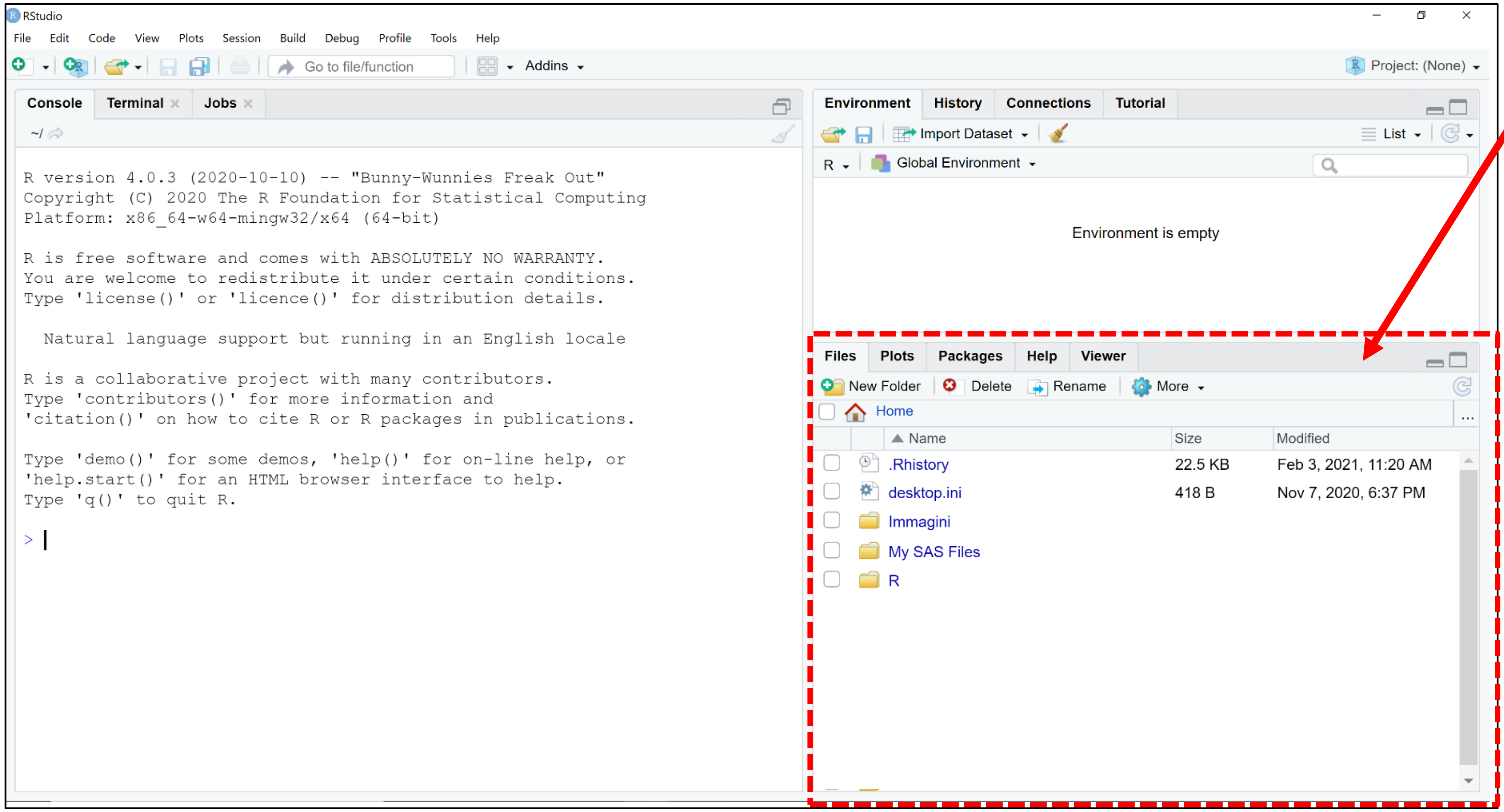


Environment/History:

In ***Environment*** information is displayed on the R objects, variables, and data currently loaded into the R workspace during a session.

History keeps track of all R commands you have executed in the Console during a session, making it easy to recall and reuse previous commands.

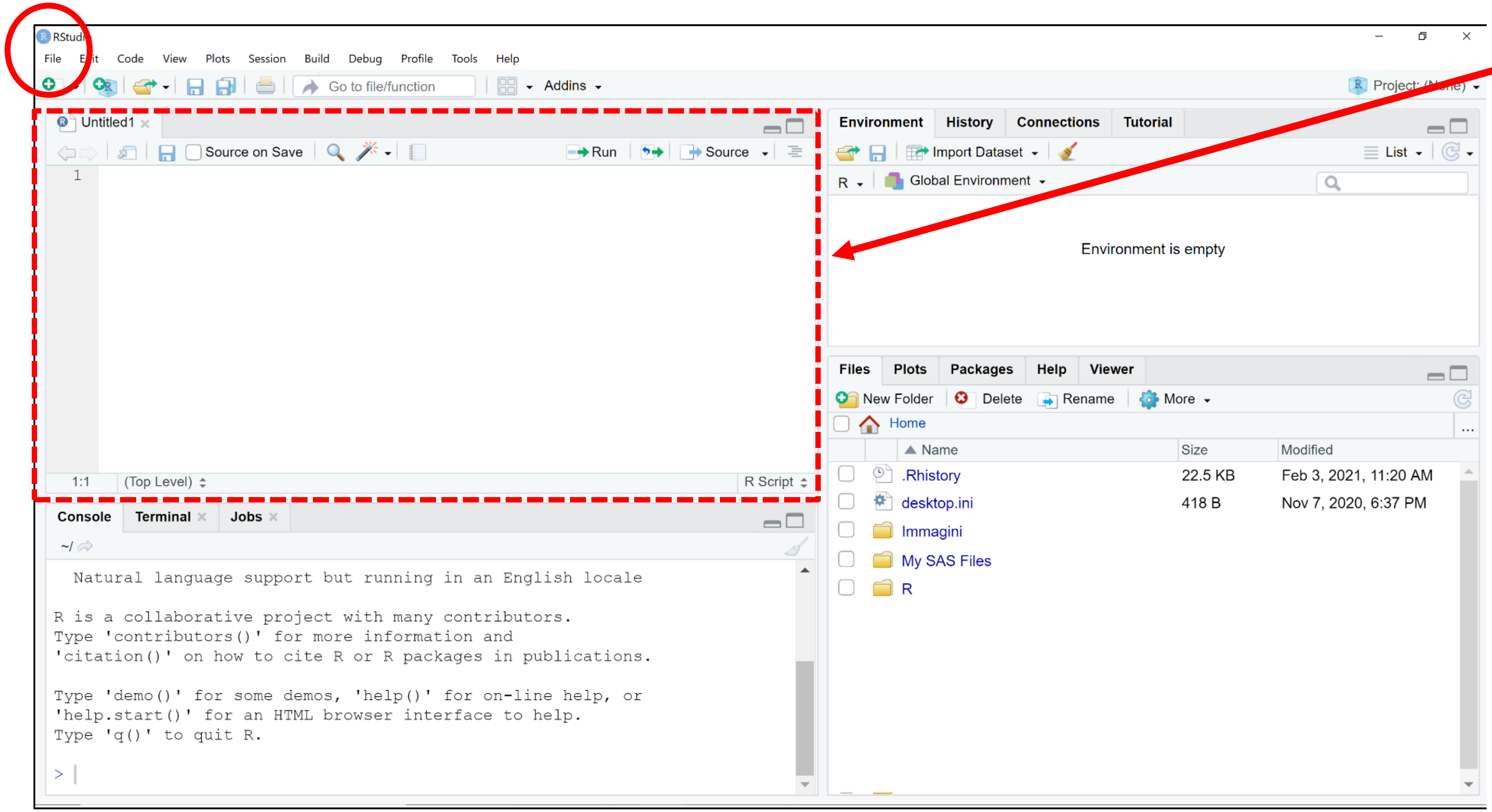
Panes in RStudio



Files/Plots/Packages/Help/Viewer: **multi-purpose** pane showing the content of a directory on disk (***Files***), any plots obtained during the session (***Plots***), The loaded and available packages (***Packages***; from this window it is also possible to install packages), and ***Help*** pages regarding functions or packages (if required)

Panes in RStudio

Selecting **NewFile|R Script** from the ***File*** menu opens the **Source pane,** which can also be opened when opening (**Open File** item on the ***File*** menu) a previously saved R script



In the **Source** pane you can write and edit R scripts. R scripts are a collection of commands written in the R programming language. To execute commands in the script, you must send them to the **Console** by selecting the portion of code to execute and pressing Enter or by clicking on the **Run** button or by placing the cursor at the end of a command and pressing Enter.

Scripts can be saved using the Save or Save as options on the *File* menu.